Phase 1A:

```
rohit@ROHIT228:~/os-u4-orange-problem$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaf
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Phase 1B:

```
rohit@ROHIT228:~/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
rohit@ROHIT228:~/os-u4-orange-problem$
```

Phase 2A:

```
rohit@ROHIT228@ROHIT228:~/os-u4-orange-problem$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
rohit@ROHIT228@ ROHITO
rohit@ROHIT228@ ROHIT228:1E90FF ~/os-u4-orange-problem$
```

Phase 2B:

```
rohit@ROHIT228:~/PES1UG24AM228-pes-vcs$ ./find_commit .git/objects/89/5aaa3637a0865c5d8f7a528c1be2acd2b | head -20
00000000: 7472 2565 2039 3100 3130 3436 3434 3434 2064 tree 91.90644 f
00000010: aabb ccdd eeff 0011 2233 4455 ... [clear minimal text]
00000020: 5045 5331 5547 3234 414d 3232 38 ... PES1UG24AM228
00000040: ...  22de 576d b9eb ba3b 8430 2070 8337 ... c.%W...40000
00000050: fe08 k03119804 0140 f142 8f32 ad6a fe62 rc...Zo...iNq...b
00000060: ...                                    (.g
rohit@ROHIT228:~/PES1UG24AM228-pes-vcs$
```

Phase 3A:

```
rohit@ROHIT228:~/os-u4-orange-problem$ ./pes init
echo "hello" > file1.txt
./pes add file1.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.h
  untracked:  commit.c
  untracked:  test_tree
  untracked:  test.txt
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  tree.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  README.md
  untracked:  index.c
  untracked:  file2.txt
  untracked:  object.c
  untracked:  tree.h
  untracked:  commit.h
  untracked:  pes.c
```

Phase 3B:

```
rohit@ROHIT228:~/os-u4-orange-problem$ ./pes add file2.txt
rohit@ROHIT228:~/os-u4-orange-problem$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.h
  untracked:  commit.c
  untracked:  test_tree
  untracked:  test.txt
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  tree.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  README.md
  untracked:  index.c
  untracked:  object.c
  untracked:  tree.h
  untracked:  commit.h
  untracked:  pes.c

rohit@ROHIT228:~/os-u4-orange-problem$
```

```
rohit@ROHIT228:~/os-u4-orange-problem$ cat .pes/index
100644 e67ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc 1776759034 8 bye.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776758453 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e719f35f9f7e65d6f606199f7f  1776757868 6 file2.txt
100644 05c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43 1776759034 12 hello.txt
rohit@ROHIT228:~/os-u4-orange-problem$
```

Phase 4A:

```
rohit@ROHIT228:~/os-u4-orange-problem$ ./pes log
commit f6538d73877b8ca76c658e56f1eafcd9f97dd82f18349c8a1d4c62043af430de
Author: PES User <pes@localhost>
Date:   1776759317

    Add farewell

commit 6cf7ede2e5a11cb127b0bb62cd6e8f3925dc53913256b97df9b786ed11e0cf8e
Author: PES User <pes@localhost>
Date:   1776759317

    Add world

commit 12f912cdd23e7b4f77f919adadb031c48446c137e9e2fd2922e0a614c3049f1d28
Author: PES User <pes@localhost>
Date:   1776759317

    Initial commit

rohit@ROHIT228:~/os-u4-orange-problem$
```

Phase 4B:

```
rohit@ROHIT228:~/os-u4-orange-problem$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/12/f912cdd23e7b4f7f919adadb031c4846c137e9e2fd2922e0a614c3049f1d28
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6c/f7ede2e5a11cb127b0bb62cd6e8f3925dc53913256b97df9b786ed11e0cf8e
.pes/objects/94/4be1b9cbcf39b4de3ce97629261c20ae59d62c055c6b986a92acf9de63fdfa
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070133357d6968b9
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f4/119ab0e6733d85f46fab31c68077da71689d1c085e1a53211275ee09218096
.pes/objects/f6/538d73877bbca76c658e56f1eafcd9f97dd82f18349c8a1d4c62043af430de
.pes/refs/heads/main
rohit@ROHIT228:~/os-u4-orange-problem$
```

Phase 4C:

```
rohit@ROHIT228:~/os-u4-orange-problem$ cat .pes/HEAD
cat .pes/refs/heads/main
ref: refs/heads/main
f6538d73877b8ca76c658e56f1eafc01156f1eafcd9f97dd82f18349c8a1d4c62043af430
rohit@ROHIT228:~/os-u4-orange-problem$
```